Diabetes Workflow Comparison

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Comparison of three regression workflows on the Diabetes dataset.

MODELS

Model 1

LinearRegression [baseline]

Description	Linear Regression evaluated on a standard train/test split.
Type	Regression
Data	Test
Tuned	None
Params	13

Model 2

RandomForestRegressor

Description	Random Forest evaluated with 5-fold cross-validation.
Type	Regression
Data	5-fold CV
Tuned	None
Params	18

Model 3

KNeighborsRegressor

Description	KNN regressor tuned with GridSearchCV and evaluated with 5-fold CV.
Type	Regression
Data	5-fold CV
Tuned	2 params
Params	16

METRICS

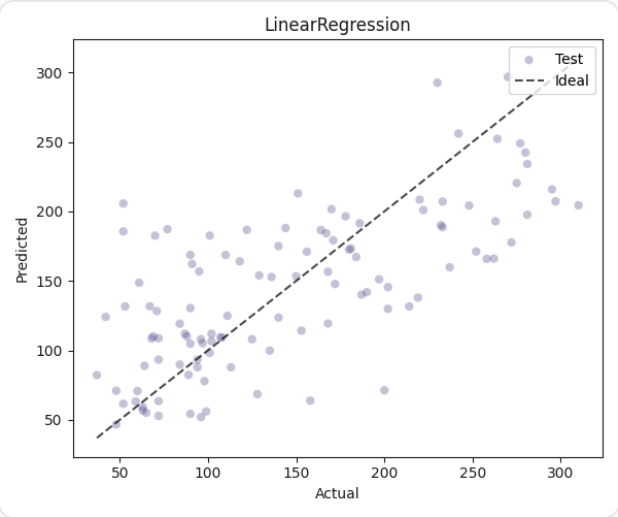
Metric	Model 1	Model 2	Model 3	Best

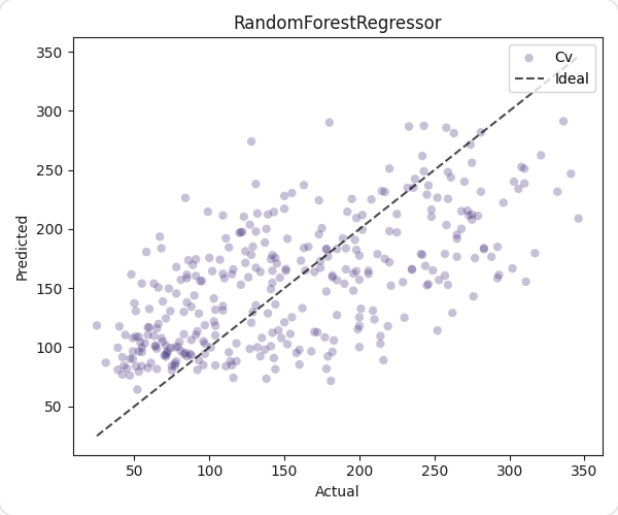
Mean Absolute Error	41.5485	49.7012 (+8.1527)	49.3585 (+7.8100)	Model 2
Max Error	153.6572	142.7364 (-10.9208)	155.8950 (+2.2378)	Model 3
Median Absolute Error	35.2108	44.9407 (+9.7299)	42.9693 (+7.7585)	Model 2
Mean Squared Error	2848.3107	3637.4399 (+789.1292)	3790.9658 (+942.6551)	Model 3
R^2 Score	0.4849	0.3763 (-0.1086)	0.3535 (-0.1314)	Model 1
Root Mean Squared Error	53.3696	60.2124 (+6.8429)	61.4718 (+8.1023)	Model 3

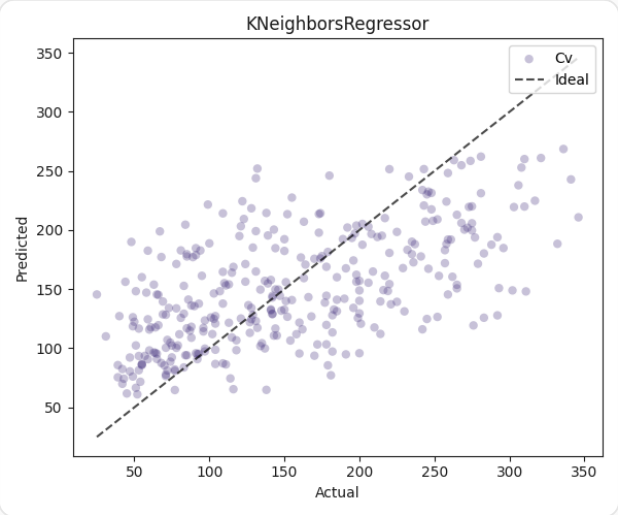
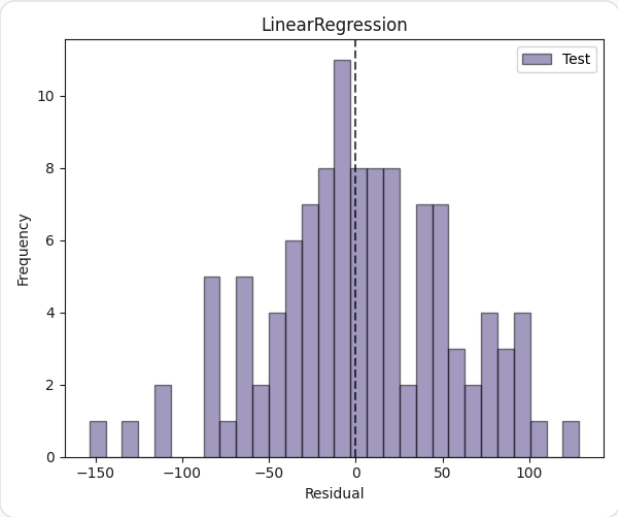
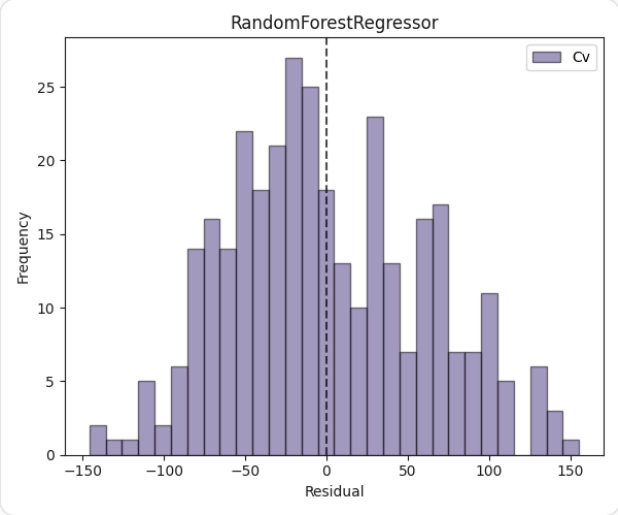
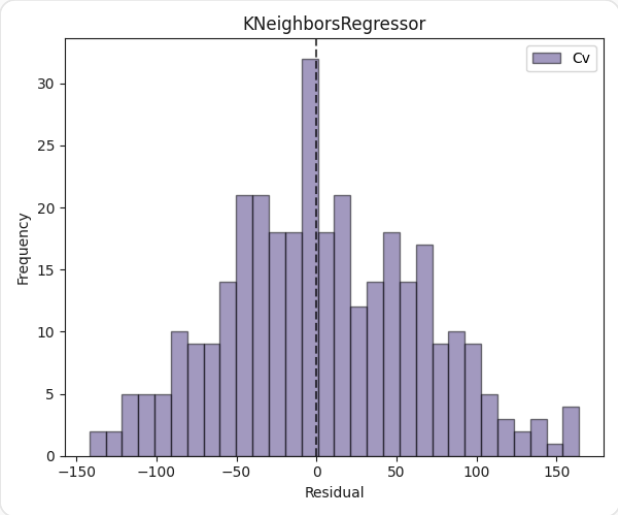
Metrics may be drawn from different evaluation splits across models.

VISUALIZATIONS

Predicted vs Actual







Residual Distribution